Lab Assignment 6

MTH 308: Numerical Analysis and Scientific Computing–I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: March 13, 2026

This lab will have you implementing the Bisection method and fixed-point iteration to approximate solutions for several problems. Please write a single script file named `Lab6LastnameRollNumber.m` and publish it as a PDF. Then submit the script `Lab6LastnameRollNumber.m` along with the corresponding published file `Lab6LastnameRollNumber.pdf` (for example, my submitted files would be `Lab6Biswas260102.m` and `Lab6Biswas260102.pdf`) to HELLO IITK.

Each solution should:

- Be contained in a separate cell, including the problem number and a short problem description.
- Be adequately commented to explain your logic.
- Run independently of other cells.

As part of your solution for each problem, output the error tolerance, the approximation, and the number of iterations required, formatted using the `fprintf` function as the sample output below:

```
Tolerance:  10e-8, Approximation:  1.23456789, Iterations:  23
```

For a solution accurate to within 10^{-k}, include at least k digits in the output.

1. Use the Bisection method to find a solution accurate to within $\varepsilon = 10^{-8}$ for $x - 2^{-x} = 0$ on the interval $[0, 1]$. Set the maximum number of iterations to be 30, and use the stopping criteria
$$\frac{b_k - a_k}{2} < \varepsilon.$$

2. Repeat Problem 1 using $\varepsilon = 10^{-12}$.

3. Plot the graphs of $y = x$ and $y = 2\sin x$. Use the Bisection method to find an approximation to within $\varepsilon = 10^{-8}$ to the first positive value of x with $x = 2\sin x$. Use the stopping criteria
$$\left|\frac{x_k - x_{k-1}}{x_k}\right| < \varepsilon.$$

4. The following four fixed-point iteration methods are proposed to compute $21^{1/3}$:

   (a) $x_k = \frac{20x_{k-1}+21/x_{k-1}^2}{21}$

   (b) $x_k = x_{k-1} - \frac{x_{k-1}^3-21}{3x_{k-1}^2}$

   (c) $x_k = x_{k-1} - \frac{x_{k-1}^4-21x_{k-1}}{x_{k-1}^2-21}$

   (d) $x_k = \left(\frac{21}{x_{k-1}}\right)^{1/2}$

   For each method, compute approximations for the value using $x_0 = 1$ and $\epsilon = 10^{-10}$ for the stopping criteria $|x_k - x_{k-1}| < \epsilon$. Rank the methods in order based on the apparent speed of convergence.